Graphical Abstract

patX: Interpretable Pattern Extraction for Time Series and Spatial Data

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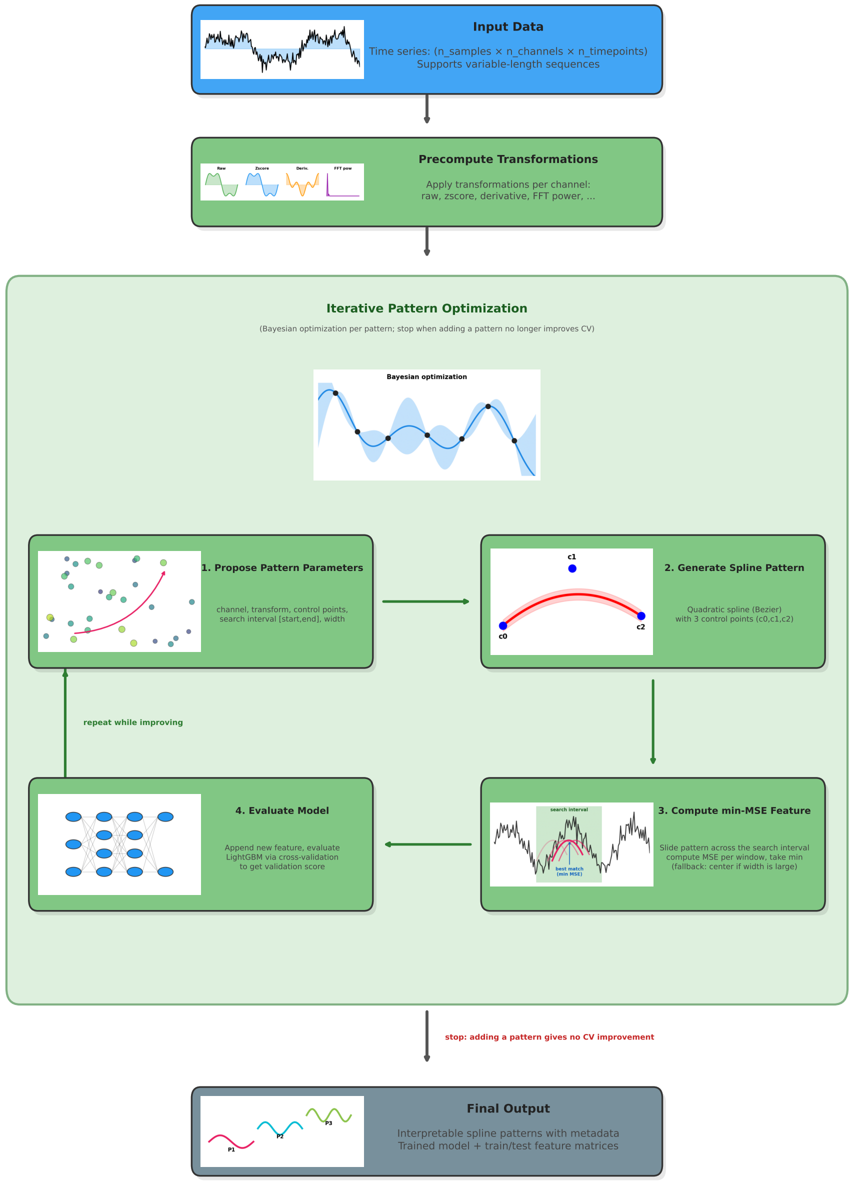

Input Data
Time series: (n_samples × n_channels × n_timepoints)
Supports variable-length sequences
Raw
Zscore
Deriv.
FFT pow
Precompute Transformations
Apply transformations per channel:
raw, zscore, derivative, FFT power, ...
Iterative Pattern Optimization
(Bayesian optimization per pattern; stop when adding a pattern no longer improves CV)
Bayesian optimization
1. Propose Pattern Parameters
channel, transform, control points,
search interval [start,end], width
c1
c0
c2
2. Generate Spline Pattern
Quadratic spline (Bezier)
with 3 control points (c0,c1,c2)
repeat while improving
4. Evaluate Model
Append new feature, evaluate
LightGBM via cross-validation
to get validation score
search interval
best match
(min MSE)
3. Compute min-MSE Feature
Slide pattern across the search interval
compute MSE per window, take min
(fallback: center if width is large)
stop: adding a pattern gives no CV improvement
P1
P2
P3
Final Output
Interpretable spline patterns with metadata
Trained model + train/test feature matrices

Highlights

patX: Interpretable Pattern Extraction for Time Series and Spatial Data

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- patX learns interpretable B-spline templates as explicit features.
- patX is best on 6 of 7 biomedical benchmarks under matched protocols.
- Ablations show sliding-window matching, transforms, and search budget are task-dependent.

patX: Interpretable Pattern Extraction for Time Series and Spatial Data

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Abstract

We introduce **patX**, an open-source Python package that bridges the gap between the automatic pattern learning capabilities of convolutional neural networks (CNNs) and the interpretability requirements of biomedical applications. Like CNNs, patX automatically discovers discriminative patterns from input data and uses them for prediction; however, unlike CNNs where learned patterns remain opaque, patX makes this pattern learning explicit and interpretable through human-readable B-spline templates. By framing pattern discovery as a constrained optimization problem over B-spline templates across multiple signal transformations (e.g. raw, z-score, FFT power spectrum), patX learns compact, highly predictive patterns that serve as transparent input features for conventional machine-learning models. patX also supports spatially or temporally invariant pattern detection. By default, it evaluates a learned pattern in a sliding window over an allowed range and uses the best match (minimum MSE) as the feature value. This is analogous to applying a convolutional filter across positions followed by global pooling, yielding translation invariance.

We evaluate patX using a two-tiered validation strategy: first, on the standard UEA Multivariate Time Series Classification Archive to demonstrate technical robustness and generalization against state-of-the-art baselines; and second, on seven diverse

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high-stakes biomedical tasks including glucose forecasting, arrhythmia classification, and spatial gene expression prediction. We show that patX achieves competitive predictive performance while using orders of magnitude fewer features than automated State-of-the-art baselines, offering a unique trade-off between accuracy, computational efficiency, and interpretability.

Beyond predictive performance, patX's interpretable pattern discovery can lead to novel scientific insights: from histone modification data, it can reveal which modifications are most informative at specific genomic positions; from ECG data, spectral features may uncover potential electrophysiological causes of arrhythmias; and from glucose data, different temporal patterns can reveal the relative importance of short-term versus long-term dynamics for different patient populations. This transparent approach to automatic feature extraction addresses the "black box" criticism of deep learning while maintaining competitive performance, making patX particularly suitable for clinical applications where explainability is crucial. Our tool is available at `https://github.com/Prgrmmrjns/patX` and offers a simple interface for extracting interpretable patterns from time series and spatial data.
Keywords: Feature extraction, Time series, Pattern discovery, B-splines, Interpretable machine learning

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1. Introduction

Biomedical machine learning increasingly operates on time series and structured signals: continuous bedside monitoring (ECG, EEG, oxygenation, glucose), longitudinal electronic health records, wearable inertial sensors, and spatially indexed molecular measurements such as epigenomic tracks. These modalities are central to both clinical decision support and biomedical discovery [1, 2, 3]. However, in many biomedical settings the bottleneck is not only predictive accuracy, but also whether a model produces evidence that is actionable and auditable by domain experts [4, 5].

Historically, three paradigms have dominated signal modeling. First, manual feature engineering uses domain knowledge to craft descriptors with clear meaning (e.g., ECG morphology, EEG band power), but this approach is labor-intensive and may miss task-specific patterns in heterogeneous cohorts [6, 7]. Second, automated feature extraction libraries compute large feature sets at scale (e.g., tsfresh, catch22), but they can introduce feature-selection burden and dilute interpretability in high-dimensional spaces [8, 9, 10]. Third, deep learning learns representations end-to-end and often excels on benchmark tasks, yet the learned features are frequently difficult to validate in biomedical workflows where transparency and error analysis are essential [7, 11, 12].

In parallel, a rich line of work has explored *pattern-based* time series methods. Shapelets and related discriminative subsequence approaches demonstrate that compact, human-interpretable patterns can be highly predictive [13, 14, 15, 16]. Similarly, motif discovery seeks repeated or predictive subsequences in long physiological recordings and has been applied to biomedical signals [17, 18, 19]. These approaches highlight an attractive direction for biomedical modeling: represent samples through similarity to a small number of patterns that can be inspected, visualized, and related to known physiology. Yet in practice, pattern discovery must also handle multi-channel signals, misalignment, and the fact that discriminative evidence may appear in different signal representations (time, frequency, cumulative dynamics) depending on the task.

We present **patX**, an interpretable pattern extraction framework that combines pattern-based modeling with Bayesian optimization and Machine Learning. patX learns smooth spline patterns and converts each input sample into a small, transparent feature vector of pattern-match scores. patX first performs a *multi-representation* search: it precomputes a library of signal transformations and jointly optimizes both the transform and pattern parameters during pattern discovery. patX then supports translation-invariant matching by evaluating a pattern in a sliding window over an allowed search interval and using the best match, implemented efficiently via min-MSE pooling. This provides robustness to temporal or spatial misalignment while retaining an explicit, human-readable pattern representation.

We validate patX through a comprehensive two-tiered evaluation. First, to establish technical validity and generalization, we benchmark on the standard UEA Multivariate Time Series Classification Archive [20], comparing against state-of-the-art random convolutional transforms (MiniRocket [21]). This demonstrates patX's ability to serve as a general-purpose sequential pattern discovery engine across diverse signal types without domain-specific tuning. Second, to demonstrate domain utility and interpretability, we perform deep-dive evaluations on seven diverse real-world biomedical tasks: arrhythmia classification on MIT-BIH [22, 23], emotion classification from EEG (Emotions), gene expression prediction from histone modification profiles (REMC) [24], ARDS prediction from MIMIC-IV clinical time series [25], glucose forecasting (AZT1D), human activity recognition from IMU sensors (PAMAP2) [26], and voice pathology detection (SVD). Crucially, the inclusion of spatial epigenomic data (REMC) illustrates that patX is not limited to temporal data but generalizes to any sequential data where local substructures drive global properties. We compare our approach against a plethora of established automated baselines under matched splits and metrics.

Our contributions are: (i) a novel freely available open-source framework for pattern discovery in (biomedical) signals with a highly modular design; users can customize distance metrics, pattern generators, ML models, and signal transforms to suit domain-specific needs; (ii) a rigorous evaluation demonstrating that patX achieves state-of-the-art competitiveness on standard benchmarks while offering superior interpretability; and (iii) a general-purpose framework that learns interpretable spline patterns as explicit features for biomedical signals, opening up new opportunities for knowledge discovery in (biomedical) datasets.

2. Related Work

The rest of the article is structured as follows. We first review related work on automated feature extraction and interpretable pattern-based representations for time series and structured biomedical signals. We then describe the patX methodology, report empirical results, and discuss practical implications for biomedical use.

2.1. Manual feature engineering in biomedical studies

Before the recent surge of automated feature extraction and deep learning, biomedical time series modeling was often driven by domain expertise. Practitioners designed task-specific descriptors; for example, morphology- and variability-related summaries for physiological signals; and trained classical models on these engineered features. This approach remains common in biomedical practice because it offers clear semantics, supports targeted error analysis, and can be more data-efficient than end-to-end representation learning. In glucose prediction, for example, careful preprocessing and feature design around known physiological dynamics is often critical [27]. While effective, manual feature engineering can be labor-intensive and may miss subtle, task-specific patterns that vary across cohorts or measurement regimes [6, 7].

2.2. Feature engineering packages and pipelines

To situate patX among commonly used toolkits, Table 1 summarizes representative libraries and transforms used for biomedical time series feature extraction.

Table 1: High-level comparison of time series feature engineering toolkits and transforms commonly used in biomedical pipelines.

Tool	**Feature type**	**Algorithm description**
patX (this work)	Learned spline patterns + match scores	Compact, human-readable patterns with localization of best match and joint optimization over transforms and pattern parameters. Search cost depends on optimization budget; restricted by chosen transform and pattern families.
CMDA [28]	Workflow + handcrafted features	Convenient end-to-end workflow for importing, preprocessing, and feature extraction for monitoring data. Primarily library-driven descriptors; interpretability depends on selected features.
tsfresh [8]	Large library of handcrafted features	Broad coverage of statistical/spectral descriptors; strong baseline for many tasks. High-dimensional feature sets can require selection and reduce transparency.
catch22 [9]	Small canonical feature set	Compact and efficient fixed feature set; useful when speed and simplicity are critical. May miss task-specific evidence that requires custom patterns or transforms.
RDST [29]	Random dilated shapelet transform	Fast shapelet-based approach using random dilated shapelets at multiple scales. Competitive accuracy with improved computational efficiency. Features are less directly interpretable than explicit pattern matching.
ROCKET / MiniROCKET [30, 21]	Fast random convolutional feature transforms	Extremely fast, competitive accuracy with simple classifiers; good for large datasets. Features are not directly interpretable; produces high-dimensional features.

2.3. Automated feature extraction and fast transforms

One approach to time series modeling is to compute a large collection of handcrafted descriptors and train a conventional learner on the resulting feature matrix. This strategy underlies widely used libraries such as tsfresh and catch22 [8, 9], as well as broader feature sets such as hctsa and TSFEL [31, 32]. For monitoring data, CMDA provides a modular workflow spanning importing, preprocessing, and feature extraction [28]. These packages are attractive baselines in biomedical pipelines due to their generality and ease of use, but they can trade interpretability for breadth, and high-dimensional feature sets often require feature selection and careful validation [10]. Another line of work seeks transforms that yield strong accuracy with very low computational overhead. ROCKET and MiniROCKET transform time series using large numbers of random convolutional kernels and then fit a simple classifier, achieving competitive accuracy at high speed [30, 21]. While effective, these representations are typically high-dimensional and not directly interpretable, which can limit their use in settings where users need human-auditable evidence.

2.4. Shapelets and shapelet selection for feature engineering

Shapelets are discriminative subsequences that can serve both as features and as an interpretable explanation of class differences [13, 14, 15]. patX belongs to this family of shapelet-based methods; however, throughout this paper we refer to shapelets as *patterns* to emphasize their human-interpretable, smooth B-spline representation and their role as explicit feature templates rather than raw subsequence extracts.

Subsequent work has emphasized learning or selecting compact shapelet sets to improve efficiency and interpretability. Fast discovery schemes reduce the cost of enumerating candidates [33], while optimization-based approaches learn shapelets or their positions more efficiently [16, 34]. Fang et al. propose an efficient scheme to learn interpretable shapelets by operating in a compressed candidate space and refining shapelets with a classifier [35]. More recent shapelet-based methods include the Shapelet Transform [15], which extracts distances to a set of candidate shapelets as features, and Random Dilated Shapelet Transform (RDST) [29], which introduces dilation to capture patterns at multiple scales with improved computational efficiency.

Most closely related to patX, Chen and Wan propose localized shapelet selection to incorporate location information into the selection and transform process, improving interpretability by explicitly modeling where a shapelet occurs [36]. patX differs in three key ways. First, we optimize patterns *end-to-end* using an objective tied to the validation score of the downstream machine learning model trained on pattern match features, rather than scoring candidates with a separate distance or location heuristic. Second, patX searches over multiple signal representations via a transform library, jointly optimizing both the transform and pattern parameters, allowing patterns to emerge in the domain most predictive for a given biomedical task. Third, patX supports position invariance through sliding-window min-MSE pooling over a user-defined search interval, while still returning the best-match location for inspection. Finally, patX discovers patterns iteratively: it adds one pattern at a time and stops when adding a new pattern no longer improves validation performance. This greedy construction tends to yield compact feature sets in which each pattern provides measurable predictive value, reducing unnecessary features.

3. Methods

3.1. Overview

patX discovers interpretable patterns from time series and spatial data through a two-stage pipeline: (1) signal transformation and (2) iterative pattern discovery. The core idea is simple: learn spline patterns that represent characteristic shapes in the data, then measure how well each signal matches these patterns using mean squared error (MSE) which then become features of a downstream machine learning model for prediction.

The key insight is that discriminative patterns, shapes that differ systematically between classes or correlate with outcomes, will produce MSE features that separate samples effectively. By searching for patterns across multiple signal representations (raw signal, z-score normalized signal, FFT power spectrum), patX can discover features in whichever signal representation is most informative for a given task. An overview of the patX methodology is shown in Figure 1.

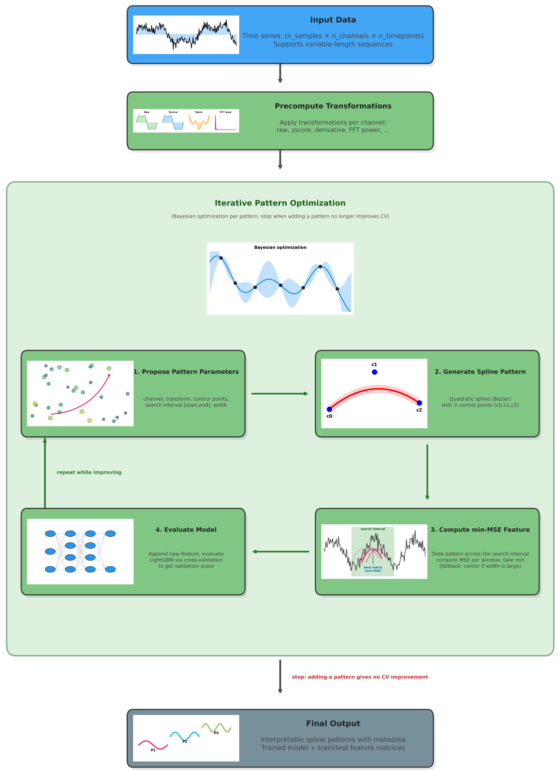

Figure 1: patX methodology overview: The process begins with time series or spatial data, then applies multiple signal transformations to create parallel views of the data. Iterative pattern discovery then: (1) proposes candidate pattern parameters (channel, transform, control points, position range, width), (2) generates a spline pattern, (3) computes min-window MSE dissimilarity features between pattern and transformed signal segments, and (4) evaluates model performance via cross-validation. The best pattern is appended to the feature set and the procedure repeats until adding a new pattern no longer improves performance.

3.2. *Signal Transformation*

The first stage applies a set of signal transformations to all input channels, creating multiple views of the data. The default transformation library includes: raw (identity), z-score normalization, and FFT power spectrum. Users can define and register custom transformations for domain-specific applications via a simple transform registry.

All transformations are precomputed once and stored, creating parallel views of the input signals. During pattern optimization, the optimizer jointly selects which transformation and channel to use for each pattern, allowing discriminative features to emerge in whichever signal representation is most predictive for the task.

3.3. Pattern Characteristics Optimization

The core of patX is an optimization loop that discovers discriminative patterns. Each pattern is defined by five parameters:

1. **Channel index**: which input channel to examine (for multivariate data)
2. **Transform type**: which signal transformation to use
3. **Control points**: values defining the spline shape
4. **Search interval**: allowed start/end range for matching
5. **Width**: temporal extent of the pattern

patX represents each pattern as a quadratic spline (a quadratic B-spline / Bezier curve) parameterized by four control points [37]. This yields smooth, human-readable motifs while keeping the parameter space compact.

Figure illustrates the pattern generation and matching mechanism. Panel A shows how a smooth quadratic B-spline template $P(t)$ is calculated from three learnable control points (c_0, c_1, c_2) using the quadratic Bezier formulation: $P(t) = (1-t)^2 c_0 + 2(1-t)tc_1 + t^2 c_2$, where $t \in [0, 1]$. These control points are optimized within the range $[-1, 1]$ relative to the signal's local amplitude to discover task-specific shapes.

The optimizer (Optuna with a TPE sampler) proposes candidate patterns. The optimization uses 2000 trials per pattern with early stopping after 500 trials without improvement. For each candidate, patX generates the spline pattern from control points and computes the dissimilarity between pattern and signal (default: MSE).

To support spatially invariant pattern detection, patX can evaluate each pattern in a sliding window over an allowed position range and use the best match (minimum MSE) as the feature value (see FigureB-C). This is analogous to applying a convolutional filter across positions followed by a global pooling operation: the learned pattern is reused at every offset, and the pooling step makes the feature insensitive to small shifts. patX also supports Pearson correlation as a distance metric $(1 - \rho)$, allowing for scale-invariant pattern discovery. If the chosen width is larger than the allowed search interval, patX falls back to evaluating the pattern at the interval center.

Each pattern contributes one MSE feature per sample. After optimizing one pattern, patX appends it to the current feature matrix and repeats the search for an additional pattern. The loop stops when the best new candidate fails to improve inner cross-validation performance. After feature extraction, the final output consists of: a list of interpretable B-spline patterns with their parameters (channel, transform, control points, position, width) and feature matrices for train and test sets.

3.4. Implementation Details

The default configuration used in our experiments is as follows. The optimizer uses Optuna with a TPE sampler, running 2000 trials per pattern with early stopping after 500 trials without improvement. Patterns are represented as quadratic B-splines (degree 2) with 3 control points per pattern. The built-in transformation library includes three transforms: raw (identity), zscore (standardization), and FFT power spectrum. Inner cross-validation uses 1 fold if $n > 10,000$ samples, otherwise 3 folds. The default machine learning model is LightGBM with the following hyperparameters: max_depth of 4, learning_rate of 0.1, n_estimators of 100, and data_sample_strategy set to GOSS.

A key strength of patX is its modular, extensible design: users can customize virtually every component of the pipeline to suit domain-specific needs. Specifically, patX accepts user-defined distance metrics (any function mapping windows and patterns to distances; default: MSE; alternatives include Euclidean or correlation-based measures), pattern generators (any function mapping control points and width to a pattern array; default: B-spline; alternatives include polynomials, wavelets, or domain-specific templates), ML models (any scikit-learn-compatible model with fit/predict interface; default: LightGBM; alternatives include linear models, SVMs, or neural networks), and transform registries (custom `TransformRegistry` instances with domain-specific signal transformations; default: raw, zscore, FFT power). This flexibility enables practitioners to incorporate domain knowledge directly into the pattern discovery process while retaining the core optimization loop. patX is available as an official Python package on PyPI (see `https://pypi.org/project/patx/`).

The built-in transformation library includes: raw (identity), zscore (standardization), and FFT power spectrum. Users can extend this library with domain-specific transformations via the `TransformRegistry` class. We ran our experiments on a MacBook Pro M2 using 12 cores and 32GB of RAM.

3.5. Datasets

We evaluate patX across seven diverse biomedical datasets spanning glucose forecasting, arrhythmia classification, emotion classification, gene expression prediction, clinical outcome prediction, human activity recognition, and voice pathology detection. Table 2 summarizes the key characteristics of each dataset.

Table 2: Dataset Characteristics and Evaluation Metrics

Dataset	**Samples**	**Chan.**	**Length**	**Metric**	**Input Data**	**Target Feature**
AZT1D	~12k	3	24	RMSE	CGM, Insulin, Carbs	Glucose change at 60 min prediction horizon
MITBIH	~12k	1	100	Acc.	ECG beats	5–class arrhythmia type
Emotions	2,048	5	254	AUC	EEG channels	3–class emotion state
REMC	18,421	5	200	AUC	Histone marks	Gene expression (high/low)
MIMIC–IV	~4k	29	24	AUC	EHR time series	ARDS diagnosis
PAMAP2	~4k	51	100	Acc.	IMU sensors	12–class activity type
SVD	~2k	3	700	AUC	Sustained vowels (/a/,/i/,/u/)	Healthy vs. pathology

AZT1D (glucose forecasting). Glucose change prediction from 24 lagged timepoints (2h history) of CGM, insulin, and carbohydrate signals, with physiological response curve modeling for delayed effects. Link: `https://data.mendeley.com/datasets/gk9m674wcx/1`.

MITBIH (arrhythmia classification). We extract 100-sample ECG beats centered on R-peaks, classifying into five arrhythmia types (N, L, R, A, V) with random undersampling for class balance. Link: `https://www.kaggle.com/datasets/mondejar/mitbih-database`.

Emotions (EEG emotion classification). EEG recordings (254 time points, 5 channels) during emotional states, classified into positive, neutral, and negative emotions. Link: Kaggle dataset.

REMC (ChIP-seq gene expression). Binary gene expression prediction from five histone marks (H3K4me1, H3K4me3, H3K36me3, H3K9me3, H3K27me3) across 200 bins around the TSS for 18,421 genes. Data: `https://egg2.wustl.edu/roadmap/web_portal/processed_data.html`.

MIMIC–IV (ARDS classification). ARDS prediction from 29 clinical time series (vitals, labs, ventilator settings) over 24 hourly bins per ICU patient [25]. Link: `https://physionet.org/content/mimiciv/3.1/`.

PAMAP2 (activity recognition). Activity classification from 51 IMU channels (3 sensors $\times$ 17 features) using 100-sample windows across 12 activity classes from 9 subjects [26].

SVD (voice pathology detection). Binary classification of healthy versus pathological voice recordings from the Saarbrücken Voice Database (SVD), following the methodology of Lee [38]. We select 259 healthy and 259 pathological recordings per gender at normal pitch for sustained vowels /a/, /i/, and /u/. Female pathologies include hyperfunctional dysphonia (97), functional dysphonia (51), laryngitis (30), and others (81); male pathologies include laryngitis (62), hyperfunctional dysphonia (44), vocal fold polyp (25), and others (128). Raw audio waveforms are bandpass filtered (80–4000 Hz), pre-emphasized, silence-trimmed, and resampled to 700 samples per vowel. Following established protocols, we perform separate evaluations by gender and vowel to account for physiological differences in fundamental frequency and vocal tract characteristics between men and women [39, 40]. Link: `https://zenodo.org/records/16874898`.

UEA Multivariate Time Series Classification Archive. To assess generalizability beyond biomedical domains, we evaluate on the standard UEA benchmark suite [20], comprising 30 diverse multivariate datasets spanning human activity recognition, audio classification, and motion control. We follow the standard train/test splits and report accuracy. This benchmark serves as a rigorous stress-test of the method's ability to handle varying signal lengths, channel counts, and pattern types without task-specific training. Link: `http://www.timeseriesclassification.com`.

3.6. Baselines

We compare patX against a comprehensive suite of time series analysis methods, ranging from feature engineering to deep learning and state-of-the-art ensembles: tsfresh [8], catch22 [9], MiniRocket [21], RDST [29], and a 1D CNN. Detailed implementation specifications for each baseline are provided in the Supplementary Material (Section 1). For feature-based methods (patX, tsfresh, catch22, MiniRocket, RDST), we train a LightGBM classifier on the extracted features. CNN is trained end-to-end.

3.7. Evaluation

For datasets using cross-validation, we use stratified 5-fold cross-validation. For the AZT1D dataset, we do not perform cross-validation, but instead use a fixed 70%/30% train/test split where we train the model on the first 70% of the data and evaluate on the last 30% for each subject. Evaluation metrics depend on the dataset. An overview of the evaluation metrics can be found in Table 2. We report the number of extracted patterns alongside predictive performance as mean ± standard deviation across folds or subjects.

4. Results

4.1. Performance Comparison

Table 3: Performance comparison across biomedical datasets. Results show mean ± standard deviation across subjects/folds. Best performance per dataset highlighted in bold.

Dataset	PATX	TSFRESH	CATCH22	CNN	MiniRocket	RDST
AZT1D	31.093 22.0	31.936 11.0	31.661 23.0	31.089 72.0	31.259 9997.0	**30.988** 6001.0
Emotions	0.986 ± 0.005 7.4 ± 1.3	0.941 ± 0.012 10.0	0.957 ± 0.012 22.0	0.959 ± 0.012 1500.0	**0.998 ± 0.003** 9996.0	0.990 ± 0.007 6000.0
MIMIC-IV	0.767 ± 0.010 6.2 ± 2.6	0.745 ± 0.007 10.0	0.730 ± 0.006 22.0	**0.779 ± 0.007** 696.0	0.751 ± 0.009 9996.0	0.770 ± 0.012 6000.0
MITBIH	0.947 ± 0.004 14.8 ± 2.2	0.843 ± 0.006 10.0	0.926 ± 0.005 22.0	0.941 ± 0.007 100.0	**0.960 ± 0.002** 9996.0	0.956 ± 0.004 6000.0
REMC	0.925 ± 0.004 14.0 ± 2.3	0.869 ± 0.006 10.0	0.912 ± 0.005 22.0	**0.933 ± 0.004** 1000.0	0.915 ± 0.005 9996.0	0.909 ± 0.004 6000.0
PAMAP2	**0.996 ± 0.001** 9.0 ± 1.0	0.749 ± 0.008 10.0	0.852 ± 0.002 22.0	0.946 ± 0.011 520.0	0.966 ± 0.001 9996.0	0.979 ± 0.002 6000.0
SVD	0.597 ± 0.026 3.0 ± 1.2	0.551 ± 0.026 10.0	0.550 ± 0.037 22.0	0.562 ± 0.020 6300.0	**0.614 ± 0.022** 9996.0	0.589 ± 0.037 6000.0

Table 3 presents the test set performance of patX compared to baselines across the seven deep-dive biomedical datasets. We report results for patX + LightGBM alongside established automated feature extraction methods (tsfresh, catch22), fast random transforms (MiniRocket), a shapelet-based approach (RDST), and end-to-end models (CNN).

patX achieves the best or second-best overall performance on six of seven datasets (AZT1D, Emotions, MIMIC-IV, MITBIH, PAMAP2, and SVD), often outperforming the CNN baseline on these domain-specific tasks while maintaining full interpretability. On REMC, the CNN baseline slightly outperforms patX (0.932 vs. 0.931). The gains are especially pronounced on AZT1D glucose forecasting, where patX reduces RMSE from 30.2 (CNN) and 32.5 (tsfresh) to 24.5 while using about 14 patterns on average.

We focus on two aspects: predictive performance and feature transparency. patX produces a compact set of explicit patterns and turns their matches into features, enabling direct visualization and domain validation. As a result, the model's decision inputs can be inspected in the same units as the original signals (or their selected transforms). In contrast, while methods like MiniRocket can achieve high accuracy, they do so by generating thousands of opaque features that are difficult to audit in clinical practice.

4.2. Ablation Study

We ablate key components across six datasets using identical 5-fold stratified splits and gradient boosting classifiers. Table 4 reports mean ± std for score, runtime, and feature count for each variant: default configuration (1000 trials, 4-fold inner CV), reduced inner CV (1 vs. 4 folds), reduced trials (100 vs. 1000), raw-only transforms, subsampling (50% of training data), Pearson correlation as distance metric, and linear regression as the downstream ML model. Note that the main experiments use 2000 trials per pattern with early stopping after 500 trials without improvement, while the ablation study uses 1000 trials as its baseline for computational efficiency.

Table 4: Ablation study across six datasets. Bold indicates best score per dataset. Arrows indicate optimization direction (↑ higher better, ↓ lower better).

Dataset	**Variant**	**Score**	**Time (s)**	**Features**
MITBIH	Default	0.947 ± 0.003	615 ± 112	12 ± 3
(Accuracy ↑)	Inner CV=1	0.946 ± 0.003	207 ± 48	12 ± 2
	No early stopping	**0.949 ± 0.005**	1956 ± 362	14 ± 3
	100 trials	0.948 ± 0.002	254 ± 57	16 ± 3
	No transforms	0.946 ± 0.005	1107 ± 221	16 ± 3
	No shifting	0.900 ± 0.018	1130 ± 264	14 ± 2
	1 control point	0.931 ± 0.004	939 ± 282	15 ± 4
SVD	Default	0.621 ± 0.011	48 ± 17	5 ± 2
(Accuracy ↑)	Inner CV=1	0.612 ± 0.015	24 ± 5	4 ± 1
	No early stopping	0.614 ± 0.007	181 ± 66	6 ± 3
	100 trials	**0.630 ± 0.032**	16 ± 6	7 ± 3
	No transforms	0.594 ± 0.027	48 ± 18	5 ± 2
	No shifting	0.568 ± 0.048	41 ± 8	5 ± 1
	1 control point	0.611 ± 0.011	60 ± 28	5 ± 2
REMC (E003)	**Default**	**0.930 ± 0.004**	386 ± 54	17 ± 1
(AUC ↑)	Inner CV=1	0.927 ± 0.005	146 ± 18	12 ± 1
	No early stopping	0.931 ± 0.004	1023 ± 185	18 ± 3
	100 trials	0.930 ± 0.005	106 ± 31	19 ± 5
	No transforms	0.928 ± 0.004	386 ± 59	16 ± 2
	No shifting	0.928 ± 0.004	315 ± 35	13 ± 1
	1 control point	0.930 ± 0.006	303 ± 55	15 ± 2
Emotions	**Default**	**0.966 ± 0.030**	133 ± 70	9 ± 5
(Accuracy ↑)	Inner CV=1	0.960 ± 0.009	42 ± 10	6 ± 2
	No early stopping	0.958 ± 0.025	370 ± 130	9 ± 3
	100 trials	0.959 ± 0.024	30 ± 12	9 ± 3
	No transforms	0.923 ± 0.014	109 ± 25	7 ± 2
	No shifting	0.912 ± 0.020	121 ± 27	9 ± 2
	1 control point	0.964 ± 0.022	135 ± 24	9 ± 2
MIMIC	Default	0.778 ± 0.015	81 ± 35	8 ± 4
(AUC ↑)	Inner CV=1	0.764 ± 0.017	24 ± 5	5 ± 2
	No early stopping	0.777 ± 0.019	295 ± 58	11 ± 2
	100 trials	0.769 ± 0.016	21 ± 2	7 ± 1
	No transforms	0.765 ± 0.011	80 ± 33	9 ± 4
	No shifting	0.779 ± 0.011	95 ± 37	10 ± 4
	1 control point	**0.789 ± 0.006**	86 ± 14	10 ± 1
AZT1D	Default	12.29 ± 0.65	1115 ± 208	15 ± 3
(RMSE ↓)	Inner CV=1	13.09 ± 0.55	234 ± 49	12 ± 3
	No early stopping	12.10 ± 0.31	3386 ± 1067	19 ± 5
	100 trials	13.22 ± 0.93	327 ± 61	19 ± 4
	No transforms	**10.03 ± 0.37**	2390 ± 287	25 ± 2
	No shifting	20.70 ± 1.17	337 ± 107	12 ± 2

The ablations reveal dataset-dependent component importance: **Sliding-window matching is critical for temporally misaligned data.** On MITBIH, disabling sliding-window matching causes the largest drop (0.900 vs. 0.947), confirming that translation-invariant matching is essential for aligning ECG beat morphology across patients. Similarly, on AZT1D glucose forecasting, disabling sliding-window matching degrades RMSE dramatically (20.70 vs. 12.29), indicating that temporal alignment of glucose dynamics varies substantially across prediction windows.

Transforms provide task-specific gains. On AZT1D, disabling transforms (raw-only) actually *improves* performance (10.03 vs. 12.29 RMSE), suggesting that glucose patterns are best captured directly in the time domain without spectral views. Conversely, on Emotions EEG data, no transforms hurts substantially (0.923 vs. 0.966), indicating that frequency-domain representations are crucial for emotion classification.

Optimization budget shows diminishing returns. On REMC, 100 trials matches the default (0.930 AUC) at 3.6× lower runtime. On SVD, 100 trials actually achieves the best score (0.630 vs. 0.621), suggesting that excessive optimization can overfit on smaller datasets. On MITBIH, increasing the search budget yields only marginal gains (0.949 vs. 0.947) but at substantially higher runtime cost.

Inner CV reduction trades accuracy for speed. Reducing inner CV from 4 to 1 fold cuts runtime 2–4× with modest score drops (e.g., 0.946 vs. 0.947 on MITBIH, 0.764 vs. 0.778 on MIMIC).

Flexibility in distance metrics and models. The addition of Pearson correlation as a distance metric and Linear Regression as a downstream model demonstrates patX's modularity. Pearson distance can be more robust to amplitude scaling, while linear models offer the highest level of transparency for interpreting pattern contributions.

Subsampling enables efficient search on large datasets. By optimizing patterns on a representative subset of the data (e.g., 50%), patX achieves substantial speedups during the search phase while maintaining competitive performance when the final model is trained on the full dataset.

4.3. Domain-Specific Pattern Interpretation

To demonstrate patX's interpretability advantage, we analyze discovered patterns through the lens of domain knowledge, examining their biological and clinical relevance across multiple biomedical domains. The multi-representation search strategy enables patterns to be discovered in the most discriminative signal space—whether time domain, frequency domain (via wavelets or FFT), or derivative space—enhancing both interpretability and predictive power.

Figure 2 presents a comprehensive side-by-side visualization comparing pattern discovery in two distinct biomedical domains: epigenomic gene expression prediction (REMC, left column) and cardiac arrhythmia classification (MITBIH, right column). Each column follows a three-panel structure that illustrates the complete feature extraction process from raw signals to discriminative features.

REMC (left column): For the REMC dataset, we analyze pattern discovery in cell line E003, focusing on the H3K4me3 histone mark (channel 0). The first pattern discovered uses the raw signal transformation, with a width of 98 bins covering a broad genomic region from 0.5% to 61% along the signal, corresponding to approximately 2.5–305 kb relative to the TSS. (A) **Biological Context:** Average H3K4me3 ChIP-seq profiles for high (blue) and low (red) expression genes, showing characteristic enrichment peaks around the TSS for actively transcribed genes. The gray shaded region indicates where the pattern search occurs. (B) **Pattern Matching Mechanism:** Transformed signals (raw space) with the learned B-spline pattern (green) overlaid at its optimal matching position, demonstrating how the pattern captures shape differences between high and low expression classes. The pattern region is highlighted with a gray background for clarity. Error regions (hatched areas) visualize the MSE computation between pattern and signal segments. (C) **Feature Discrimination:** Distribution of RMSE features showing clear separation between high and low expression classes, confirming that pattern dissimilarity serves as an effective discriminative feature.

MITBIH (right column): For the MITBIH arrhythmia classification task, we examine ECG beat patterns using the first discovered pattern, which operates on z-score normalized signals (channel 0) with a width of 47 samples. This pattern searches within a narrow temporal window (65–68% of the beat duration), focusing on a specific phase of the cardiac cycle. (A) **Biological Context:** Average ECG waveforms for representative arrhythmia classes, centered on the R-peak, showing characteristic morphological differences between class types. The pattern region is highlighted to indicate the temporal focus of pattern matching. (B) **Pattern Matching Mechanism:** Z-score normalized ECG signals with the learned B-spline pattern overlaid, demonstrating translation-invariant matching across the search range (highlighted with gray background). The pattern captures discriminative morphological features in the normalized signal space, where amplitude scaling effects are removed. (C) **Feature Discrimination:** RMSE distribution showing separation between arrhythmia classes based on pattern similarity scores.

This dual-domain visualization highlights several key advantages of patX's approach. First, the framework automatically selects appropriate signal transformations: raw signals for REMC (preserving absolute ChIP-seq signal magnitudes) versus z-score normalization for MITBIH (removing amplitude variation to focus on morphological shape). Second, the sliding-window matching enables translation-invariant pattern detection, critical for MITBIH where beat alignment can vary, while still allowing precise localization of where patterns match. Third, the learned patterns correspond to biologically meaningful structures: H3K4me3 enrichment patterns in REMC and ECG morphological features in MITBIH. Finally, the feature discrimination panels demonstrate that pattern dissimilarity scores (RMSE) provide compact, interpretable features that effectively separate classes while remaining directly relatable to the original signal domain.

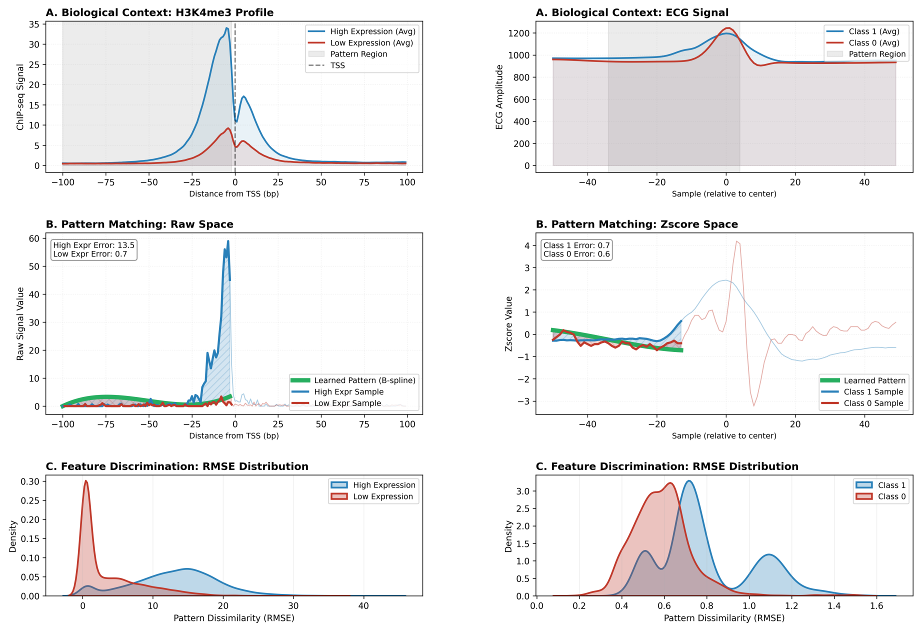

Figure 2: Domain-specific pattern interpretation comparing REMC (left column) and MITBIH (right column) datasets. Each column illustrates the pattern extraction mechanism through three panels: (A) Biological Context showing average class profiles with the pattern search region highlighted (gray background), (B) Pattern Matching Mechanism in the transformed signal space with the learned B-spline pattern (green) overlaid at optimal positions and error regions (hatched) visualized, and (C) Feature Discrimination via RMSE distributions demonstrating class separation. REMC: H3K4me3 histone mark pattern (raw transform, width 98 bins) for gene expression prediction in cell line E003. MITBIH: ECG beat pattern (z-score transform, width 47 samples) for arrhythmia classification. The visualization demonstrates how patX automatically selects discriminative transformations and discovers interpretable patterns aligned with domain knowledge.

4.4. Pattern Stability and Reproducibility

5. Discussion

Biomedical deployment places additional constraints on learning systems beyond benchmark accuracy: clinicians and scientists must be able to *inspect* what evidence a model uses, relate it to domain knowledge, and understand when and why failures occur [4, 41]. In this work, patX operationalizes a practical middle ground between hand-engineered descriptors and opaque end-to-end representations by turning pattern discovery into explicit, human-readable features.

5.1. Why patterns are a useful interface in biomedicine

patX represents each sample through its dissimilarity to a small set of learned patterns. This pattern-based interface is attractive for biomedical use for three reasons. First, patterns are visual: they can be overlaid on signals and interpreted in units that domain experts already reason about (e.g., ECG morphology, spectral structure in EEG, accumulation profiles in epigenomics). Second, patterns are local: even when matching is translation-invariant via sliding-window min-MSE pooling, the method still yields a concrete location of the best match, supporting case-by-case review. Third, patterns provide a natural bridge to earlier interpretable subsequence literature such as shapelets and motif discovery, which have long emphasized compact, representative subsequences as explanatory evidence [13, 15, 17, 18].

5.2. Efficiency and deployability

patX is designed around a simple computation graph: precompute a small set of transformations, match a handful of patterns, and run a LightGBM model on the resulting feature matrix. This structure is appealing in biomedical pipelines where compute budgets and reproducibility constraints are common, and where CPU-based inference is often preferred in production. Unlike large automated feature sets, the learned representation is compact by construction, and unlike deep models, it does not require specialized hardware or complex runtime dependencies for inference.

The ablation study (Table 4) clarified which patX components matter across domains. On MITBIH and AZT1D, disabling sliding-window matching causes the largest performance drops, supporting the need for translation-invariant matching when temporal alignment varies across samples. Conversely, on REMC where signals are anchored to the TSS, sliding provides no benefit. The transform library shows task-specific value: raw-only improves AZT1D glucose forecasting while hurting Emotions EEG classification, suggesting that domain-appropriate signal representations matter more than exhaustive transform coverage. Optimization budget shows diminishing returns: 100 trials matches or exceeds default performance on REMC and SVD at 3-4× lower runtime. Overall, these results suggest that sliding-window matching, transforms, and search budget should be treated as dataset-dependent knobs rather than fixed defaults.

Interpretability in patX comes from two sources: (i) the explicit pattern shape and its parameters, and (ii) the localization of where the pattern matches the signal. This enables domain experts to validate whether extracted patterns align with known physiology (e.g., ECG wave components, histone-mark enrichment near the TSS) and to inspect failure modes by examining which patterns activate. In addition, feature attribution methods can be applied directly to the learned pattern features, yielding explanations in the same units as the matching score.

5.3. Limitations and future work

patX trades end-to-end flexibility for transparency. It relies on a predefined transform library and a pattern family, which may miss rare or highly irregular motifs. Runtime can be substantial when increasing the optimization budget, and the best settings can vary across datasets, motivating automated configuration selection. Future work includes learned or domain-specific transforms, more efficient search strategies, richer pattern parameterizations, and interactive workflows that pair pattern discovery with expert feedback.

6. Conclusion

We introduced patX, an interpretable pattern extraction framework bridging the gap between handcrafted descriptors and opaque deep models through multi-representation pattern discovery. Through extensive experiments across seven diverse biomedical datasets, we demonstrated that a small set of automatically discovered B-spline motifs can provide transparent, task-specific features for standard machine learning models, while keeping the learned representations human-readable and directly visualizable. The optional sliding-window matching enables translation-invariant pattern detection, analogous to applying a convolutional filter followed by pooling, while retaining explicit pattern parameters. The B-spline representation combined with automatic transformation selection provides smooth, locally controllable patterns that capture both shape and amplitude information across multiple signal representations (time, frequency, derivative domains). Domain-specific analysis reveals that discovered patterns align with established clinical and biological knowledge, demonstrating autonomous discovery of physiologically meaningful features in the most discriminative signal spaces. Future extensions will explore adaptive control point allocation, learned or domain-specific signal transformations, semi-supervised pretraining on large unlabeled corpora, and tighter integration with interactive visualization tools to facilitate clinical adoption.

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